Asthma — Comprehensive Clinical Guide

1. Overview and Epidemiology

Asthma is a chronic inflammatory disease of the airways characterized by variable and recurring symptoms, airflow obstruction, bronchial hyperresponsiveness, and an underlying inflammation. It affects approximately 339 million people worldwide. In India, asthma affects around 15-20 million people.

Asthma can begin at any age but most commonly starts in childhood. It has both genetic and environmental components. Children with a family history of asthma, allergic rhinitis, eczema (atopic triad), or food allergies are at higher risk.

The global burden of asthma is significant in terms of lost school days, reduced quality of life, emergency visits, and healthcare costs. India accounts for a disproportionate share of asthma-related deaths, partly due to underdiagnosis and undertreatment.

2. Pathophysiology

The central feature of asthma is airway inflammation, predominantly eosinophilic in allergic asthma. Inhaled allergens activate mast cells and Th2 lymphocytes, triggering release of histamine, leukotrienes (LTD4, LTC4), prostaglandins, and cytokines (IL-4, IL-5, IL-13).

These mediators cause: airway smooth muscle contraction (bronchoconstriction), mucus hypersecretion, airway edema, and over time, structural changes (airway remodeling) — subepithelial fibrosis, smooth muscle hypertrophy, angiogenesis — which reduce reversibility.

Triggers activate already hyperresponsive airways: allergens (house dust mite, pollen, mold, cockroach, pet dander), respiratory infections (especially rhinovirus), air pollutants (PM2.5, ozone, diesel exhaust), tobacco smoke, exercise, cold air, strong odors, aspirin/NSAIDs, beta-blockers, and emotional stress.

3. Classification and Severity

	FEV1	SABA Use

	>80% predicted	≤2 days/week
ut not daily	>80%	>2 days/week
	60-80%	Daily
	<60%	Multiple times/

GINA (Global Initiative for Asthma) classifies asthma by level of control rather than intrinsic severity: Well-controlled (symptoms ≤2 days/week, no nighttime symptoms, no activity limitation, no exacerbations), Partly controlled, and Uncontrolled.

4. Symptoms and Signs

Classic symptoms include: episodic wheeze (musical whistling sound, especially on expiration), chest tightness, shortness of breath, and cough (particularly nocturnal or early morning). Symptoms are often worse at night or with exercise.

Physical examination: expiratory wheeze on auscultation, prolonged expiration, use of accessory muscles in moderate attacks. In a severe attack: inability to speak in full sentences, tachypnea (>30 breaths/min), tachycardia (>120 bpm), paradoxical pulse, cyanosis.

Life-threatening attack: silent chest (no wheeze despite effort — airways severely obstructed), exhaustion, bradycardia, hypotension, confusion or reduced consciousness. This is a medical emergency.

5. Diagnosis

Clinical diagnosis is supported by spirometry showing: FEV1/FVC ratio <0.7 (obstruction), and significant reversibility — ≥12% AND ≥200 ml increase in FEV1 after 400 mcg salbutamol (bronchodilator test).

Peak Expiratory Flow (PEF) monitoring: a ≥20% variation in PEF over 2 weeks (morning dip pattern) supports asthma diagnosis. Useful for self-monitoring at home.

Bronchial challenge tests (methacholine, exercise, mannitol): used when symptoms suggest asthma but spirometry is normal. A positive test confirms airway hyperresponsiveness.

Additional tests: skin prick testing or specific IgE (RAST) to identify allergen triggers. Fractional exhaled nitric oxide (FeNO) — elevated in eosinophilic airway inflammation, predicts steroid response.

6. Pharmacological Treatment

Reliever (Rescue) Medications

Short-Acting Beta-2 Agonists (SABA): salbutamol (albuterol) 100 mcg MDI — 2-4 puffs up to every 4-6 hours as needed. Rapid onset (5-10 min), peak effect 30-60 min, duration 4-6 hours. Overuse (>2 canisters/month) indicates poorly controlled asthma and increased mortality risk.

Ipratropium bromide (anticholinergic): used in acute severe asthma alongside salbutamol in hospital. Not for routine maintenance. Useful in patients who cannot tolerate SABAs.

Controller (Preventive) Medications

Inhaled Corticosteroids (ICS): budesonide, beclomethasone, fluticasone. The cornerstone of asthma management. Reduce airway inflammation, prevent exacerbations, reduce hospitalization and mortality. Must be taken daily even when symptom-free. Rinse mouth after use to prevent oral candidiasis.

Long-Acting Beta-2 Agonists (LABA): salmeterol, formoterol. Never used as monotherapy for asthma (increased risk of fatal attacks). Always used in combination with ICS (ICS/LABA combinations: budesonide/formoterol, fluticasone/salmeterol).

Leukotriene Receptor Antagonists: montelukast 10 mg daily. Useful add-on, especially in aspirin-sensitive asthma and co-existing allergic rhinitis. Well-tolerated orally.

Biologics for severe asthma: omalizumab (anti-IgE), mepolizumab (anti-IL-5), dupilumab (anti-IL-4/13). For severe uncontrolled eosinophilic or allergic asthma. High cost limits access in India.

7. GINA Stepwise Treatment Approach

	Preferred Controller	Reliever
nt)	As-needed low-dose ICS-formoterol OR none	SABA or low-dose ICS-formoterol
istent)	Low-dose ICS daily	SABA or ICS-formoterol
)	Low-dose ICS-LABA	SABA or ICS-formoterol
	Medium/high-dose ICS-LABA	SABA or ICS-formoterol
y)	Add tiotropium, biologic, or low-dose OCS	SABA

8. Acute Asthma Attack Management

Mild-Moderate Attack: Remove patient from trigger. Sit patient upright. 4 puffs salbutamol MDI + spacer every 20 minutes for 3 doses. Monitor response. If improving, reduce to 4 puffs every 3-4 hours.

Severe Attack (hospital): Continuous nebulized salbutamol + ipratropium. IV/oral corticosteroids (prednisolone 40-50 mg/day or hydrocortisone IV). Controlled oxygen (target SpO2 93-95%). IV magnesium sulfate 2g over 20 min in refractory cases.

Life-threatening: ICU admission. IV bronchodilators, IV aminophylline infusion, possible intubation and mechanical ventilation. Heliox (helium-oxygen mixture) reduces airway resistance.

Discharge criteria: PEF >75% predicted or personal best, minimal symptoms, SABA use <4-hourly, able to sleep. Prescribe ICS, provide written Asthma Action Plan, arrange follow-up within 1 week.

9. Asthma in Special Situations

Exercise-Induced Bronchoconstriction (EIB): typically peaks 5-10 minutes after stopping exercise. Prevention: 2 puffs SABA 15 minutes before exercise. Regular ICS use reduces EIB. Adequate warm-up reduces severity.

Occupational Asthma: caused by workplace sensitizers (latex, flour, isocyanates, wood dust, laboratory animals). Symptoms improve on weekends/holidays. Key management: identify the occupational cause and remove exposure as early as possible.

Asthma and pregnancy: most asthma medications are safe in pregnancy. Uncontrolled asthma is more harmful to the fetus than the medications used. ICS, SABAs, and montelukast are considered safe. Oral corticosteroids should be used when indicated.

Asthma in children: diagnosis is challenging under 5 years. Trial of ICS is both diagnostic and therapeutic. Address allergen exposure at home (no pets in bedroom, dust mite-proof covers, no indoor smoking).

10. Long-Term Management and Education

The Asthma Action Plan is a written individualized guide for patients with 3 zones: Green (controlled — maintain therapy), Yellow (worsening — increase therapy), Red (emergency — immediate action and call emergency services). Every asthma patient should have one.

Inhaler technique is a major determinant of treatment success. Poor technique leads to up to 94% of patients not receiving adequate medication. Device selection should match patient ability; spacers improve delivery from MDIs by 40-50%.

Trigger avoidance: identify personal triggers through history and allergy testing. HEPA air purifiers reduce indoor allergen load. Immunotherapy (allergy shots or sublingual drops) can modify the disease course in allergic asthma over 3-5 years.

Comorbidity management: allergic rhinitis (treat with nasal steroids — reduces asthma exacerbations), GERD (treat with PPI), obesity (weight loss improves asthma control), obstructive sleep apnea (CPAP therapy).

Annual influenza vaccine and pneumococcal vaccine (adults >65 or severe asthma) are recommended. Measure spirometry at least annually in all asthma patients. Review inhaler technique at every visit.